Status — one line per numbered result

The short answer. `PaperInventory.md` is the long one and stays as the audit trail; this file is the thing to read first.

Measured on `main` after r0047: build **1364 jobs, 0 errors, 0 warnings**, `sorry` **0**, `axiom` **0**, `sorryAx` **0**. Package: 117 modules, 43,481 lines, 2,019 theorems.

Counts

#		Count
1	Numbered results in Gunter & Scott	30
2	**— fully proved**	**24**
3	— partly proved, some part open	3
4	**— refuted as stated** (our transcription or the paper)	2
5	— never transcribed into the tree	2
6	Paper properties (numbered conjuncts + prose claims)	239
7	— stated and proved	169
8	— stated, proof open (`S+H`)	**12**
9	— stated, but not as the paper states it	18
10	— no Lean statement at all	36
11	`Prop`-valued claims nothing proves	**7** (8 strict)

Rows 8 and 11 are the two kinds of unfinished work, and they are different: row 8 is a real Lean statement with an open proof; row 11 is a `def` naming a claim nobody attempted. **`sorry` 0 sees neither.**

The 30 numbered results

`P` proved · `p` partly proved · `R` refuted as stated · – not in the tree

#	Result		Status
1	Theorem 1	`P`	least fixed point, and below every fixed point
2	Theorem 2	–	**not quoted anywhere in the tree or the docs**
3	Theorem 3	`P`	`theorem3`, `theorem3_existsUnique`
4	Lemma 4	`P`	`NormalSubposet.lean`
5	Lemma 5	`P`	`FinitaryProjection.lean`
6	Theorem 6	`P`	`theorem6`
7	Theorem 7	`p`	sentence 1 proved; sentences 2–3 proved only at the degenerate strength — **every domain has an `EffectivePresentation`** because `Classical.dec` fills its fields. The recursive forms are open: `Theorem7ArrowRecursive`, `Theorem7StrictRecursive`
8	Lemma 8	`P`	`Product.lean`
9	Lemma 9	`P`	`lem9_*` — **two printed misprints repaired** (items 3 and 5)
10	Lemma 10	`P`	7 conjuncts, `lem10_{prod,smash,sum,separated,lift,strict}`
11	Theorem 11	`P`	`thm11`, with `thm11_converse`
12	Theorem 12	`P`	`thm12` and the three powerdomains
13	Lemma 13	`P`	`lem13_smyth`, `lem13_hoare`
14	Theorem 14	`P`	`thm14`, both directions
15	—	–	**not quoted anywhere in the tree or the docs**
16	Theorem 16	`P`	`thm16`, `thm16_positive`
17	Lemma 17	`P`	10 conjuncts. r0047 **removed `[BoundedComplete β]`** from `lem17_fun` and `lem17_strictFun`
18	Theorem 18	`P`	`thm18` — closed r0042, `[propext, Classical.choice, Quot.sound]`
19	Lemma 19	`P`	`lem19`
20	Lemma 20	`P`	`lem20`
21	Theorem 21	`P`	`thm21`
22	Theorem 22	`P`	`thm22`
23	Lemma 23	`P`	`lem23`
24	Lemma 24	`P`	`lem24` (Gunter & Scott's; **not** Gunter 1987's Lemma 24, below)

#	Result		Status
25	Theorem 25	P	`thm25`, `thm25_isUniversal`
26	Theorem 26	p	`thm26` carries an added `hs : ∀ i, 0 < s i`. The justification — that the theorem is false at arity 0 — **is not established**; the argument refutes the paper's *proof*, not its statement
27	Theorem 27	P	`thm27`
28	Lemma 28	R	**refuted at generic `U`** (`not_forall_lemma28`, witness `Flat Empty`), and stays false after adding `[Domain U]` and `[BoundedComplete U]`. **Proved at `U`**: `lemma28AtU`, all nine conjuncts. The blocker is `UniversalForBCD U`
29	Theorem 29	p	sentence 1 proved (`thm29`). Sentence 2: `Thm29Second` **refuted** — our transcription dropped the paper's word "domain". `Thm29SecondAtDomains`, the true reading, is **open**
30	Lemma 30	R	universal closure **refuted** (`not_forall_lemma30`). `Lemma30AtV` is **open**, now at arity 3

Rows 2 and 15 are a gap in our record, not a measurement of the paper. Nothing in the tree or the docs quotes them. Whether Gunter & Scott number a result 2 and a result 15 has not been checked against the printed text.

What is open, in one table

#	Item	What it needs
1	`StepFunctionsDecidable`	**restate over `consistentEnum`** — the guard `IsCompactElement (ofPairs Q)` is *not* the boundedness test, kernel-checked. Machinery built. Closes 4 claims
2	`Theorem7ArrowRecursive`	item 1, then `Primrec` facts for the `Finset (ℕ × ℕ)` coding
3	`Theorem7StrictRecursive`	item 1, likewise
4	`Thm29Normal`	the `M(f)` tower refutes Theorem 25's hypothesis; the η tower is not a fixed point. Three routes named, one is refuting it outright
5	`Thm29SecondAtDomains`	item 4
6	`Lemma30AtV`	item 4, plus `FpImagesBifinite V`
7	`Lem30Arrow`	item 6
8	`PreservesRecursivePresentation`	proved at `fstOp`/`sndOp`; open at the arrow, where it is equivalent to item 2

Items 2–3 and 5–7 are downstream of 1 and 4. Only items 1 and 4 are real work, and item 1 is mechanical.

Additional theories

Results from other authors, formalized here because the paper's proofs need them. None is part of Gunter & Scott's 30.

Jung, *Cartesian Closed Categories of Domains*

#	Result	Status
1	Corollary 1.36	**proved** — `JungCor136`, not by Jung's route (which goes through Prop. 1.22 and a retraction pair) but by indexing below `cap e`
2	Theorem 1.37	**proved at `[Domain D]`** — `R45.Agent5.thm137`. Full generality open
3	Theorem 1.37 for chains	**proved at `[Domain D]`** — `thm137Chains`
4	Proposition 1.22	not needed — the route around it is item 1
5	Theorem 2.1, 2.3	quoted; five printed defects in Jung's write-up recorded

Iwamura and Markowsky

#	Result	Status
1	Iwamura's lemma	**proved** — `exists_chain_directed_cover`. **Never in Mathlib**
2	Markowsky's theorem	**proved** — `hasChainSuprema_iff_hasDirectedSuprema`. **Never in Mathlib**

#	Result	Status
3	`HasWellOrderedInfima`	**no producer exists** — five occurrences, all consumers. A complete reduction chain with an empty left end

Spreen 2005

#	Result	Status
1	Lemma 5.8	**proved** — `PropertyM.hasOmegaOpBoundsAbove_pair`. This, not Iwamura, is what closed Jung's 1.37 here

Gunter 1987, *Universal Profinite Domains*

#	Result	Status
1	Lemma 24 at `M(A)`	**proved** — `R47.Agent1.lemma24_MPair`. **The first proof anywhere**: Gunter's printed Lemma 24 produces *some* `A⁺` and is not about `M(A)`; the `M(A)` form is a remark with no theorem and no proof
2	`HasNormalRealizations` at `Ainf`	**refuted** — `not_hasNormalRealizations_Ainf`. The route to `Thm29Normal` is sound but its hypothesis is unsatisfiable at this tower
3	Proposition 21, Theorems 22 and 25	used, via the implication `thm29Normal_of_hasNormalRealizations`

Defects in the printed paper

Nine, recorded in `StatementRecovery.md`. Three suspected tenths were checked and turned out to be **our** transcription errors, not the paper's:

#	Suspected	Actually
1	Theorem 26 false at arity 0	refutes the paper's *proof*, not its statement
2	Theorem 29's second sentence false	our transcription dropped "domain" from "any bifinite domain"
3	the step-function enumeration	our guard tests compactness where the paper tests boundedness

Reproducing

```
scripts/counts.sh                  # modules, lines, theorems, sorry
scripts/numbered-status.sh         # declarations per numbered result
scripts/a6-env-scan.sh <out>       # then a6-summarize.py for row 11
scripts/a5-r47-conditional.sh      # row 8, the S+H conditional surface
scripts/compile.sh -r <round>      # build
```